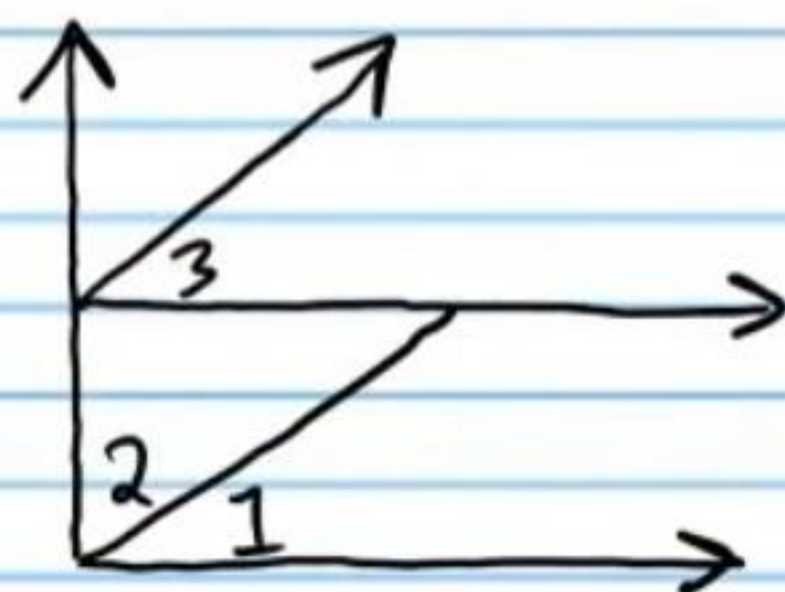


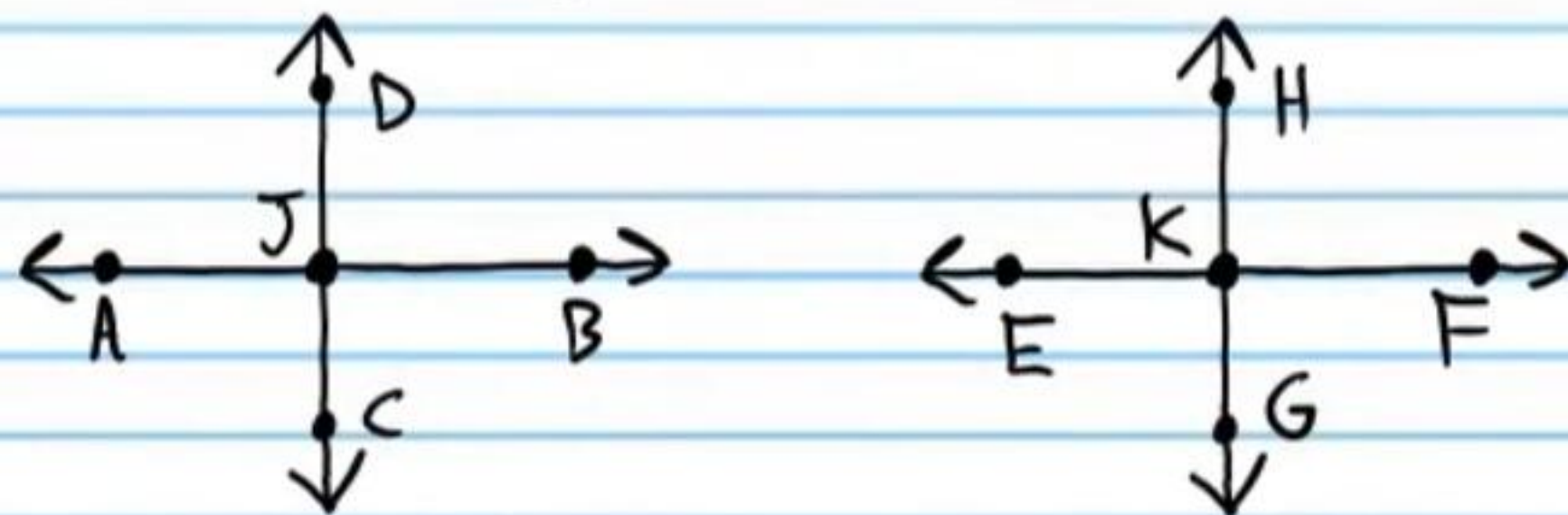
Proofs and Properties of Equality and Congruence (Homework)

Prove the following statements with two-column proofs.

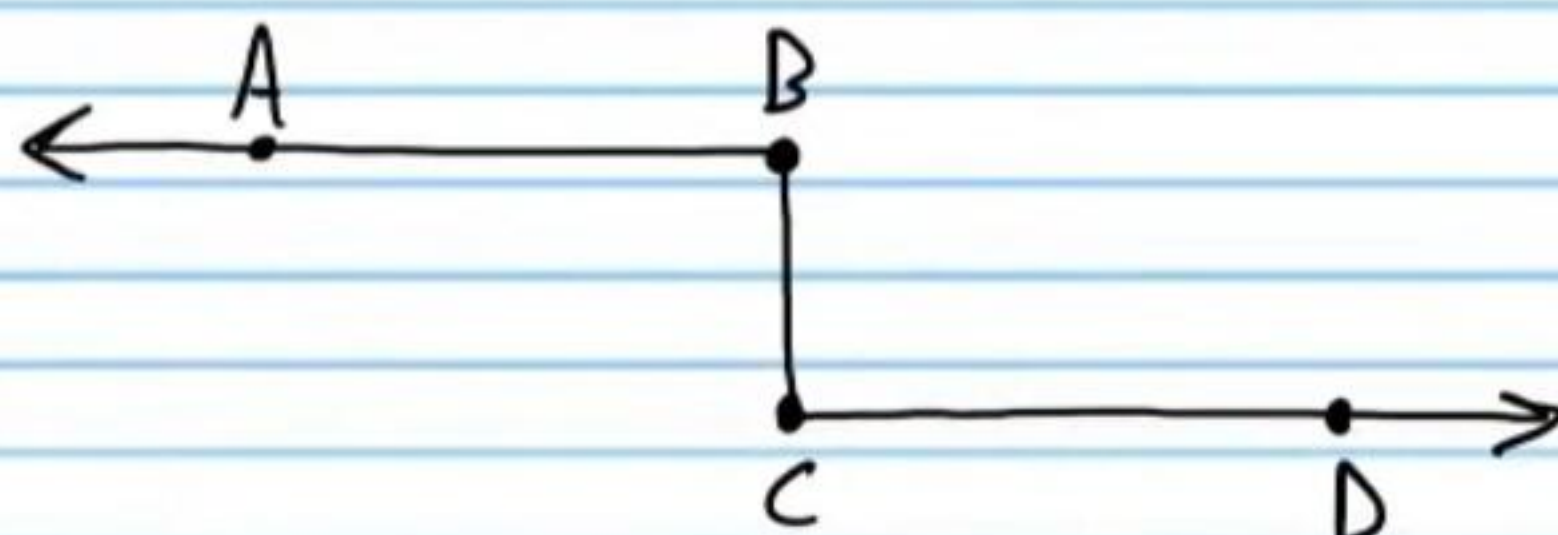
① If $\angle 1$ is complementary to $\angle 2$ in the figure below, and $\angle 3$ is complementary to $\angle 2$, then $\angle 1 \cong \angle 3$.



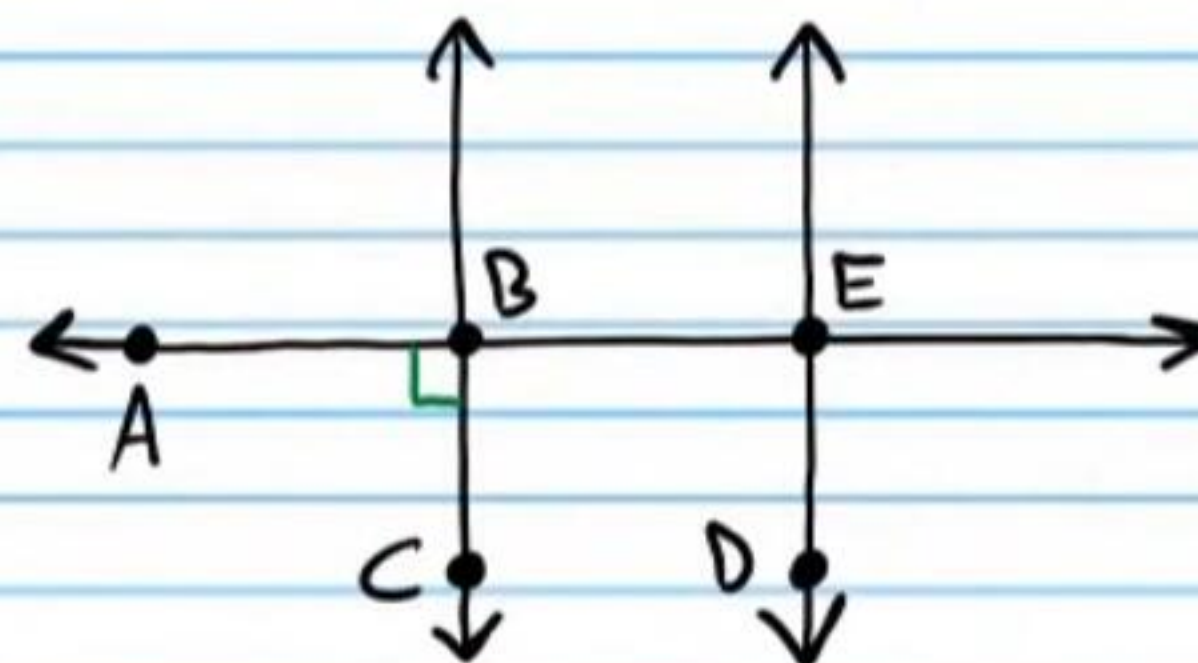
② If $\overleftrightarrow{AB}$ is perpendicular to $\overleftrightarrow{CD}$ and $\overleftrightarrow{EF}$ is perpendicular to $\overleftrightarrow{GH}$ in the figure below, then $\angle AJD \cong \angle EKG$.



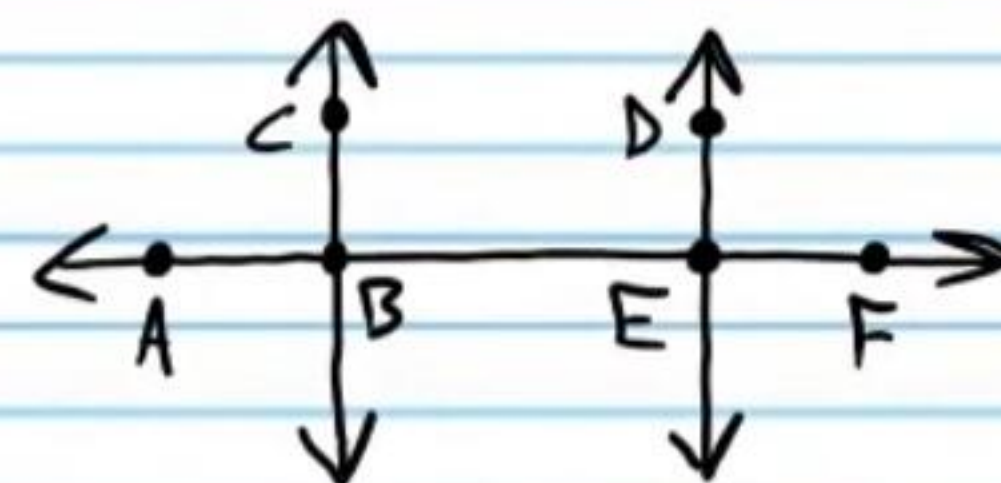
③ If $\overrightarrow{BA} \perp \overrightarrow{BC}$ below and $\overrightarrow{BC} \perp \overrightarrow{CD}$, then $\angle ABC \cong \angle BCD$.



④ If $\angle ABC \cong \angle DEB$, then $\overleftrightarrow{BE} \perp \overleftrightarrow{DE}$.

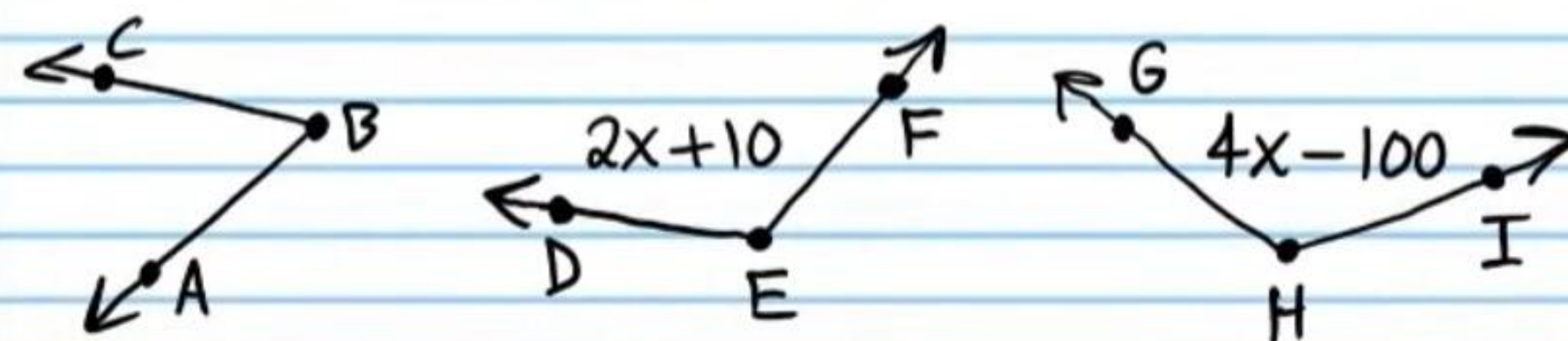


⑤ If $m\angle CBE = 90^\circ$ and $\overleftrightarrow{EF} \perp \overleftrightarrow{DE}$, then $\angle DEF \cong \angle CBE$.

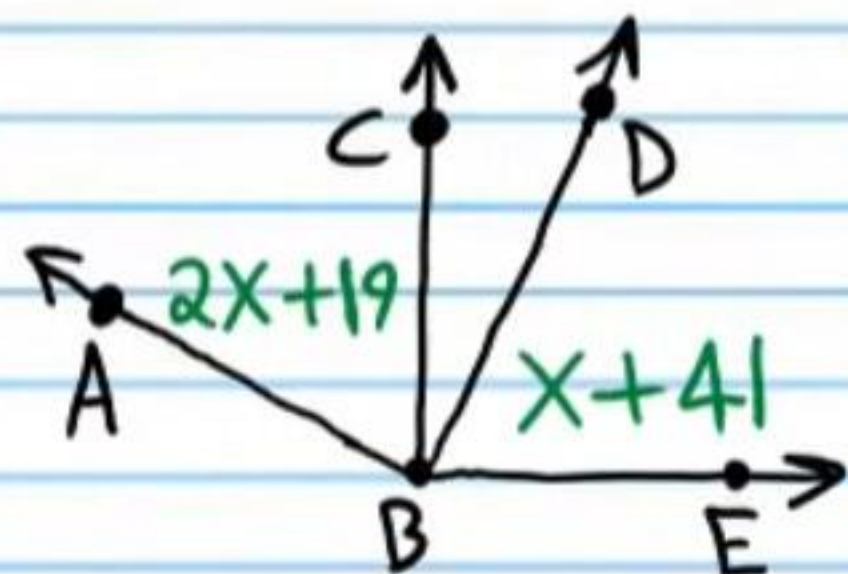


⑥ If $\angle A$ & $\angle B$ are both rotation angles (360°), then they are congruent.

⑦ If $\angle ABC$ is supplementary to $\angle DEF$ and $\angle ABC$ is supplementary to $\angle GHI$, then $x = 55$.



⑧ If $\angle ABC$ is complementary to $\angle CBD$ below and $\angle CBD$ is complementary to $\angle DBE$, then $x = 22$.



Answers for Homework

①	Statement	Reason
	I) $\angle 1$ and $\angle 2$ are complementary. $\angle 3$ and $\angle 2$ are complementary.	Given.
	II) $\angle 1 \cong \angle 3$	$\cong$ Comp. Thm.

②	Statement	Reason
	I) $\overleftrightarrow{AB} \perp \overleftrightarrow{CD}$ and $\overleftrightarrow{EF} \perp \overleftrightarrow{GH}$	Given.
	II) $\angle AJD$ and $\angle EKG$ are rt $\angle$'s	Def. of $\perp$ Lines.
	III) $\angle AJD \cong \angle EKG$	Right $\angle \cong$ Thm.

③	Statement	Reason
	I) $\overrightarrow{BA} \perp \overrightarrow{BC}$ & $\overrightarrow{BC} \perp \overrightarrow{CD}$	Given.
	II) $\angle ABC$ & $\angle BCD$ are rt. $\angle$'s.	Def of $\perp$.
	III) $\angle ABC \cong \angle BCD$	Rt. $\angle \cong$ Thm.

④	Statement	Reason
	I) $\angle ABC \cong \angle DEB$	Given.
	II) $m\angle ABC = m\angle DEB$	Def. of $\cong \angle$'s.
	III) $m\angle ABC = 90^\circ$	Given (diagram)
	IV) $90 = m\angle DEB$	Trans. Prop of $=$.
	V) $\angle DEB$ is a rt $\angle$.	Def. of rt. $\angle$.
	VI) $\overleftrightarrow{BE} \perp \overleftrightarrow{DE}$	Def. of $\perp$ lines.

⑤	Statement	Reason
	I) $\overleftrightarrow{EF} \perp \overleftrightarrow{DE}$	Given.
	II) $\angle DEF$ is a rt. $\angle$.	Def of $\perp$ lines.
	III) $m\angle CBE = 90^\circ$	Given.
	IV) $\angle CBE$ is a rt. $\angle$.	Def. of rt. $\angle$.
	V) $\angle DEF \cong \angle CBE$	Rt. $\angle \cong$ Thm.

⑥	Statement	Reason
I)	$\angle A$ & $\angle B$ are rotation $\angle$'s.	Given
II)	$m\angle A = 360^\circ$ & $m\angle B = 360^\circ$	Def. of rotation $\angle$.
III)	$m\angle A = m\angle B$	Trans. Prop of =.
IV)	$\angle A \cong \angle B$	Def. of $\cong \angle$'s.

⑦	Statement	Reason
I)	$\angle ABC$ is supplementary to $\angle DEF$ and $\angle ABC$ is supplementary to $\angle GHI$.	Given.
II)	$\angle DEF \cong \angle GHI$	$\cong$ Supp. Thm.
III)	$m\angle DEF = m\angle GHI$	Def. of $\cong \angle$'s.
IV)	$m\angle DEF = 2x + 10$ & $m\angle GHI = 4x - 100$	Given (diagram).
V)	$2x + 10 = 4x - 100$	Subs. Prop of =.
VI)	$10 = 2x - 100$	Subt. Prop of =.
VII)	$110 = 2x$	Add Prop of =.
VIII)	$55 = x$	Div Prop of =.

⑧	Statement	Reason
I)	$\angle ABC$ is complementary to $\angle CBD$ and $\angle CBD$ is complementary to $\angle DBE$.	Given.
II)	$\angle ABC \cong \angle DBE$	$\cong$ Comp. Thm.
III)	$m\angle ABC = m\angle DBE$	Def of $\cong \angle$'s.
IV)	$m\angle ABC = 2x + 19$ and $m\angle DBE = x + 41$.	Given (diagram).
V)	$2x + 19 = x + 41$	Subs. Prop of =.
VI)	$x + 19 = 41$	Subt. Prop of =.
VII)	$x = 22$	Subt. Prop of =.